User Manual

Merchant + operator guide · generated November 09, 2026

1. Install

```
python -m venv .venv
.venv\Scripts\activate   (macOS/Linux: source .venv/bin/activate)
pip install -r requirements.txt
```

2. Build data and model (one command)

```
python run_pipeline.py
```

Generates synthetic data, cleans it, engineers features, trains the model, seeds the database, runs every QA suite and rebuilds these documents.

3. Merchant dashboard

```
python 4_Backend/api/fraud_detection_api.py   → open http://127.0.0.1:5000
```

- **Top tiles**: orders scored, sent to review, blocked, money at risk, awaiting review.
- **Risk score distribution**: green = approve (0-12), amber = review (13-77), red = block (78-100). The thresholds are chosen per model on a validation period and shown in the header.
- **Flagged transactions**: highest risk first, with the reasons. Click Approve / Block / Fraud / False alarm - decisions are stored; Fraud / False alarm also set the order's label (export with connection.py --export-labels for retraining).
- **Check an order**: score any order by hand to see how the model reacts (dry run - nothing is stored).

4. Integrate with your checkout

```
POST /api/score  {"user_id": "C123", "amount": 249.0, "ip_country": "US", "billing_country":
"US",
 "is_vpn": 0, "device_id": "fp_9a1", "card_hash": "tok_77", "account_created": "2026-10-01
09:00:00",
 "checkout_seconds": 95, "login_attempts": 1, "failed_logins": 0, "pages_viewed": 6}
```

Response: risk_score, risk_level, action (approve / review / block) and reasons. Only user_id and amount are required; missing behaviour signals default to neutral values. A missing account_created is taken from the customer's first known order, or treated as unknown - never as 'created just now'.

Action	What to do
approve	Ship normally.
review	Hold shipping; verify by email/phone or ask for 3-D Secure, then record your decision.
block	Cancel/refund before shipping. Mark 'Fraud' to confirm, or 'False alarm' if it was genuine.

5. Quality-assurance console

```
python 3_Model/evaluation/run_tests.py
streamlit run 5_Frontend/streamlit_app/app.py
```

- **Evaluation**: per-test scores for LLM prompts, golden fraud cases and model gates.
- **Run history**: quality over time. **Model comparison**: champion vs baselines.
- Run with --provider openai --model gpt-4o-mini (and OPENAI_API_KEY) to evaluate a real LLM analyst; mock mode needs no key.

6. Troubleshooting

Symptom	Fix
Dashboard empty	python 4_Backend/database/connection.py --seed
FileNotFoundError fraud_model.pkl	python run_pipeline.py (or 3_Model/training/train_model.py)
Drift gate fails	Retrain on recent data; check the drifted features in the report
OPENAI_API_KEY is not set	Set it, or use --provider mock